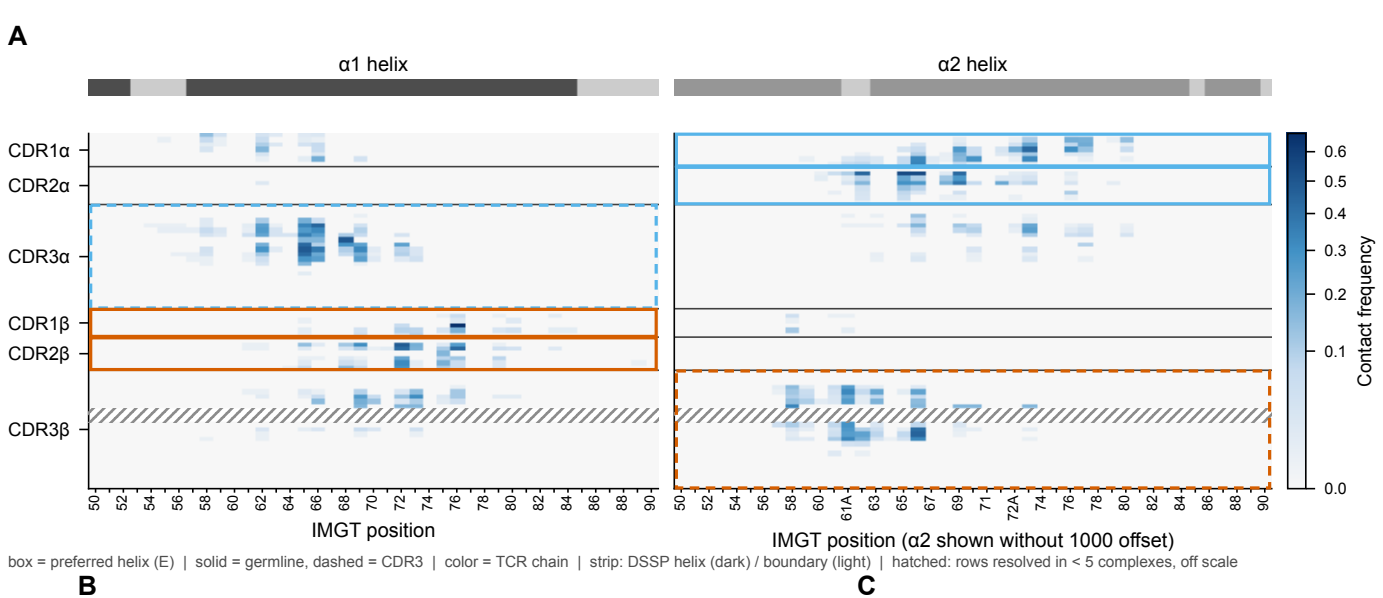
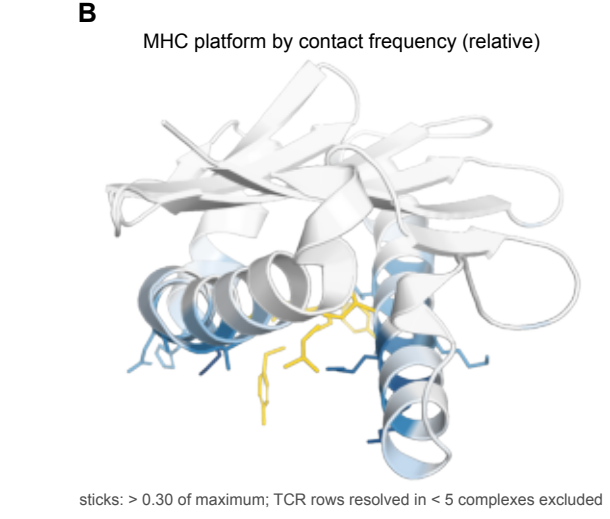
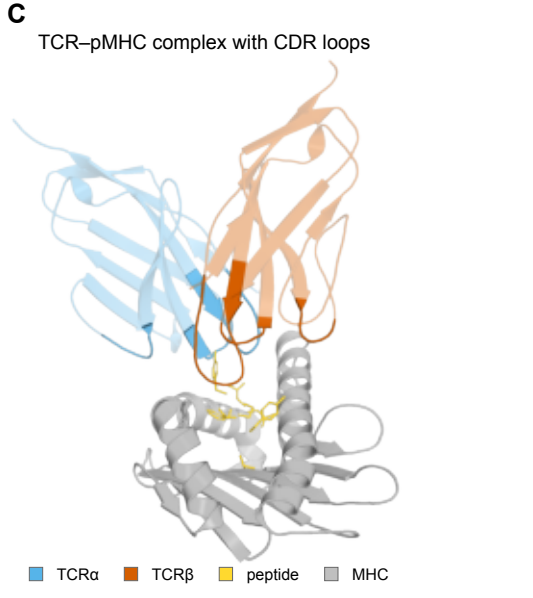
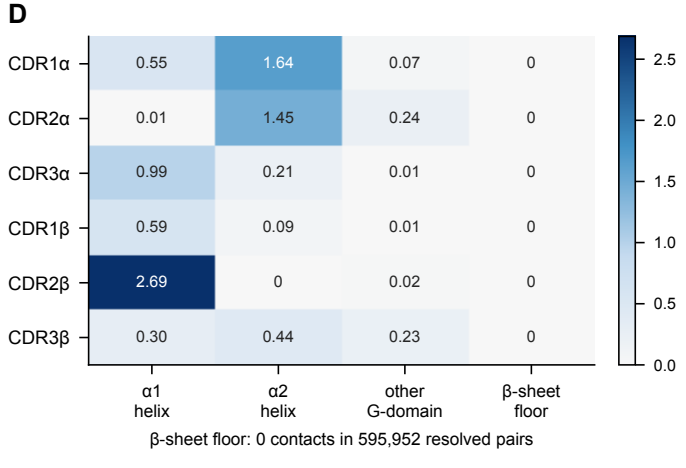
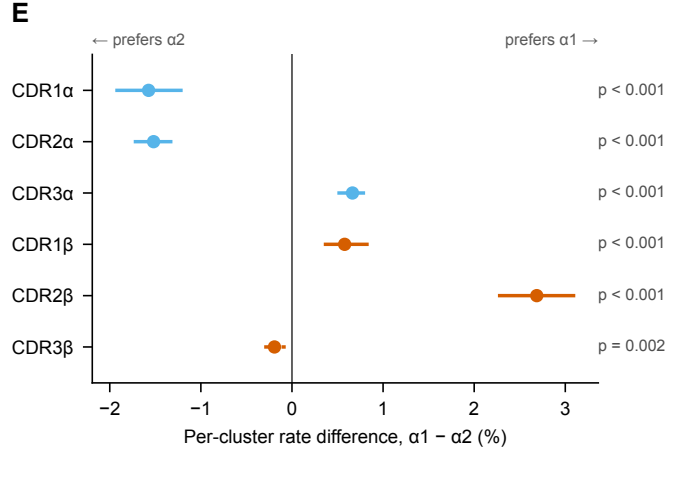

A
α1 helix
α2 helix
CDR1α
CDR2α
CDR3α
CDR1β
CDR2β
CDR3β
Contact frequency
IMGT position
IMGT position (α2 shown without 1000 offset)
box = preferred helix (E) | solid = germline, dashed = CDR3 | color = TCR chain | strip: DSSP helix (dark) / boundary (light) | hatched: rows resolved in < 5 complexes, off scale


B
MHC platform by contact frequency (relative)
sticks: > 0.30 of maximum; TCR rows resolved in < 5 complexes excluded


C
TCR–pMHC complex with CDR loops
TCRα
TCRβ
peptide
MHC


D
CDR1α 0.55 1.64 0.07 0
CDR2α 0.01 1.45 0.24 0
CDR3α 0.99 0.21 0.01 0
CDR1β 0.59 0.09 0.01 0
CDR2β 2.69 0 0.02 0
CDR3β 0.30 0.44 0.23 0
α1 helix
α2 helix
other G-domain
β-sheet floor
Contact rate (%)
β-sheet floor: 0 contacts in 595,952 resolved pairs


E
← prefers α2
prefers α1 →
CDR1α p < 0.001
CDR2α p < 0.001
CDR3α p < 0.001
CDR1β p < 0.001
CDR2β p < 0.001
CDR3β p = 0.002
Per-cluster rate difference, α1 − α2 (%)